

WEEK 1 ASSIGNMENT REPORT

Python Fundamentals for Data Analytics

Internship	Internova Internship
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Week	Week 1
Assignment	Python Fundamentals for Data Analytics
Tools Used	Python and PyCharm

Submitted as part of the Internova Data Analytics Internship.

1. Introduction

This report presents the work completed during Week 1 of the Internova Internship. The assignment introduces the fundamental concepts of Python programming required for Data Analytics. The tasks include basic syntax, variables, operators, decision-making, loops, functions, collections, file handling, and a small student management project.

2. Objectives

- Understand the basic syntax and structure of Python programs.
- Learn variables, data types, and input/output statements.
- Perform arithmetic operations using Python operators.
- Use conditional statements to make decisions.
- Apply for loops and while loops.
- Create and use simple functions.
- Work with lists and dictionaries.
- Read and write data using files.
- Build a small menu-driven Python project.

3. Tasks Completed

Task	Work Completed
Task 1: Python Basics	Displayed a welcome message and collected the student's name, college, and branch using input().
Task 2: Variables and Data Types	Created integer, float, string, and Boolean variables and displayed their data types.
Task 3: Operators	Performed addition, subtraction, multiplication, division, and modulus operations.
Task 4: Conditional Statements	Calculated a grade using if, elif, and else statements.
Task 5: Loops	Printed numbers from 1 to 20, generated a multiplication table, and displayed even numbers.
Task 6: Functions	Created a function to calculate the square of a number.
Task 7: Collections	Used lists and dictionaries to store and display student information.
Task 8: File Handling	Created a text file, wrote student information into it, and read the file content.
Task 9: Mini Project	Developed a basic Student Management System with add, display, search, and delete options.

4. Screenshots

The screenshots of all 9 tasks have been saved separately in the submitted screenshot folder. Each screenshot shows the related Python code and its output in PyCharm.

5. Tools and Technologies Used

- Python 3
- PyCharm IDE
- GitHub
- Google Drive

6. Learning Outcomes

By completing this assignment, I improved my understanding of Python programming and learned how to write simple programs using variables, conditions, loops, functions, collections, and files. I also developed a basic menu-driven Student Management System.

7. Conclusion

The Week 1 assignment provided a strong foundation in Python for Data Analytics. The concepts learned in this assignment will be useful for future work involving data processing, analysis, and automation.